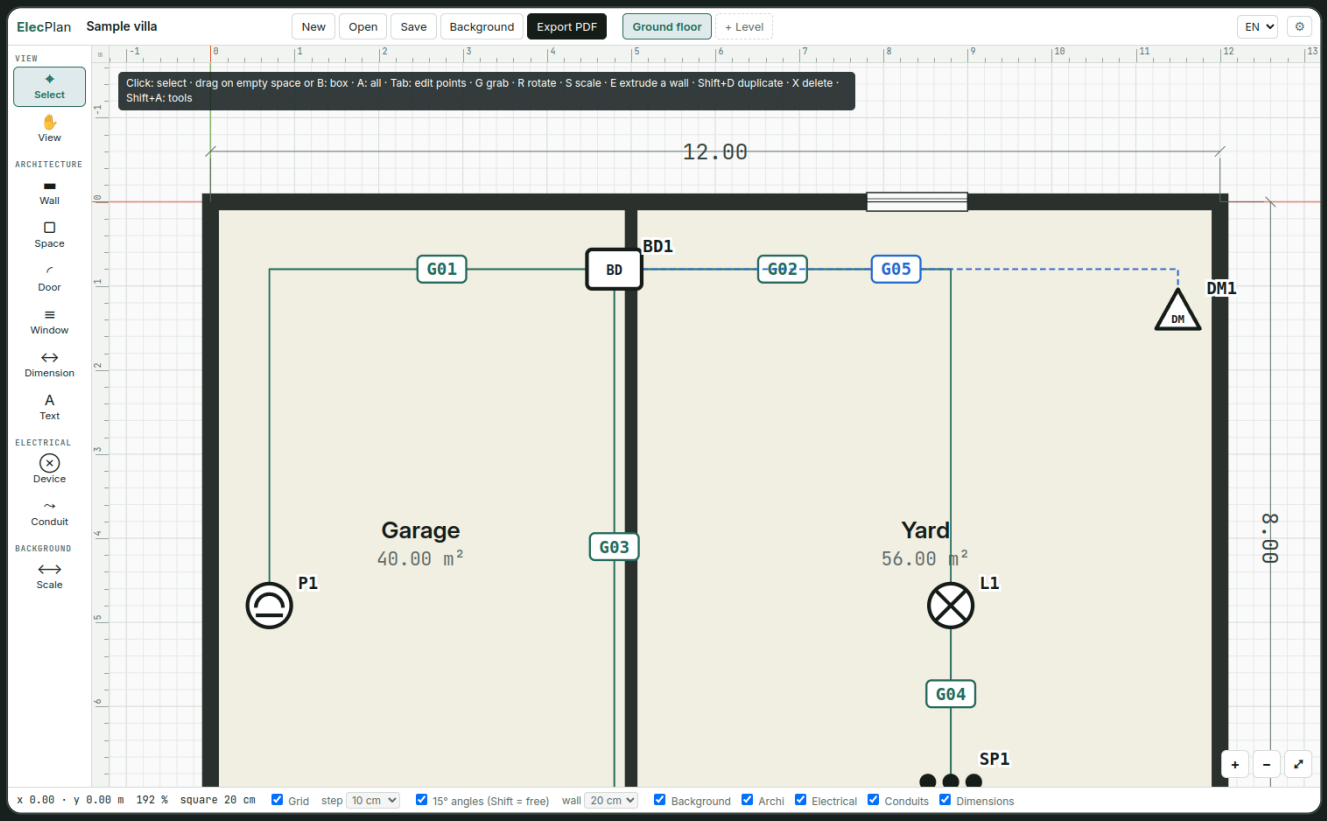
USER MANUAL · VERSION 0.1

ElecPlan

Architectural and electrical plans, with AI assistance from Claude Code.



PLATFORM
Linux (AppImage, .deb)

FILES
project .elecplan.json · A3 PDF · SVG · CSV

AI ASSISTANCE
Claude Code installed on the PC, signed in to a Claude account

EDITION
21 October 2026

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01

Overview

In short

ElecPlan draws a building plan (walls, spaces, doors, windows, dimensions) and then, on top of it, the electrical plan: referenced devices, junction boxes, numbered conduits and circuits. It produces a printable A3 plan, a bill of materials and, with the AI assistant, wiring proposals, a compliance check and a complete electrical file.

Principles

- **Unit: the centimetre.** Coordinates, thicknesses and widths are in cm; lengths are displayed in metres.
- **Levels.** A project holds several levels (ground floor, upper floor, basement…), each with its own plan and background.
- **Unique references.** Every device gets an automatic reference (P1, I2, BD3…) that is unique across the project.
- **Conduits are numbered once and for all.** G01, G02… A number is never reassigned automatically.
- **Offline.** Drawing, files and exports work without Internet. Only the AI assistant uses Claude.
- **Bilingual.** The interface is available in English or French (see *The main screen*).

The three layers of a plan

Architecture — Thick walls, spaces with their area and a movable label, doors and windows that snap to walls, dimensions, texts. A background plan (image or PDF) can be used as a tracing layer.

Electrical — 17 device types (sockets, single / two-way / ON switches, detectors, light fittings, spotlights, junction boxes, panel, enclosure, RJ45…), each with circuit, mode, height and note.

Conduits — Drawn from one device to another, with bends (straight or Bézier curves). Kind (power, low voltage, spare, outdoor), tube and content (wire codes).

The AI assistant is optional: without Claude Code, every drawing and export feature is still available.

02

Installation

Option A · Debian / Ubuntu package

```
sudo apt install ./elecplan_0.1.0_amd64.deb
```

The application appears in the menu as **ElecPlan**. To uninstall it: `sudo apt remove elecplan`.

Option B · portable AppImage

```
chmod +x ElecPlan-0.1.0.AppImage
./ElecPlan-0.1.0.AppImage
```

No installation: the file runs as is, from a USB stick for example. Some recent distributions need `libfuse2` installed to run AppImages.

Where is my data?

ITEM	LOCATION
Projects	wherever you save them (`.elecplan.json` files; older `.fpp.json` files open too)
Automatic recovery	the application's local storage, on every change
Settings (language, preferences, recent files, assistant)	`~/.config/ElecPlan/settings.json`
AI tasks	`~/.config/ElecPlan/jobs/<id>/`

AI assistant · Claude Code

To use the assistant, install Claude Code once (Node.js 18 or later required):

```
npm install -g @anthropic-ai/claude-code
claude
# in Claude Code: /login
```

Sign in with your Claude account (Pro or Max plan). ElecPlan then runs Claude Code in the background; it needs no key and stores none.

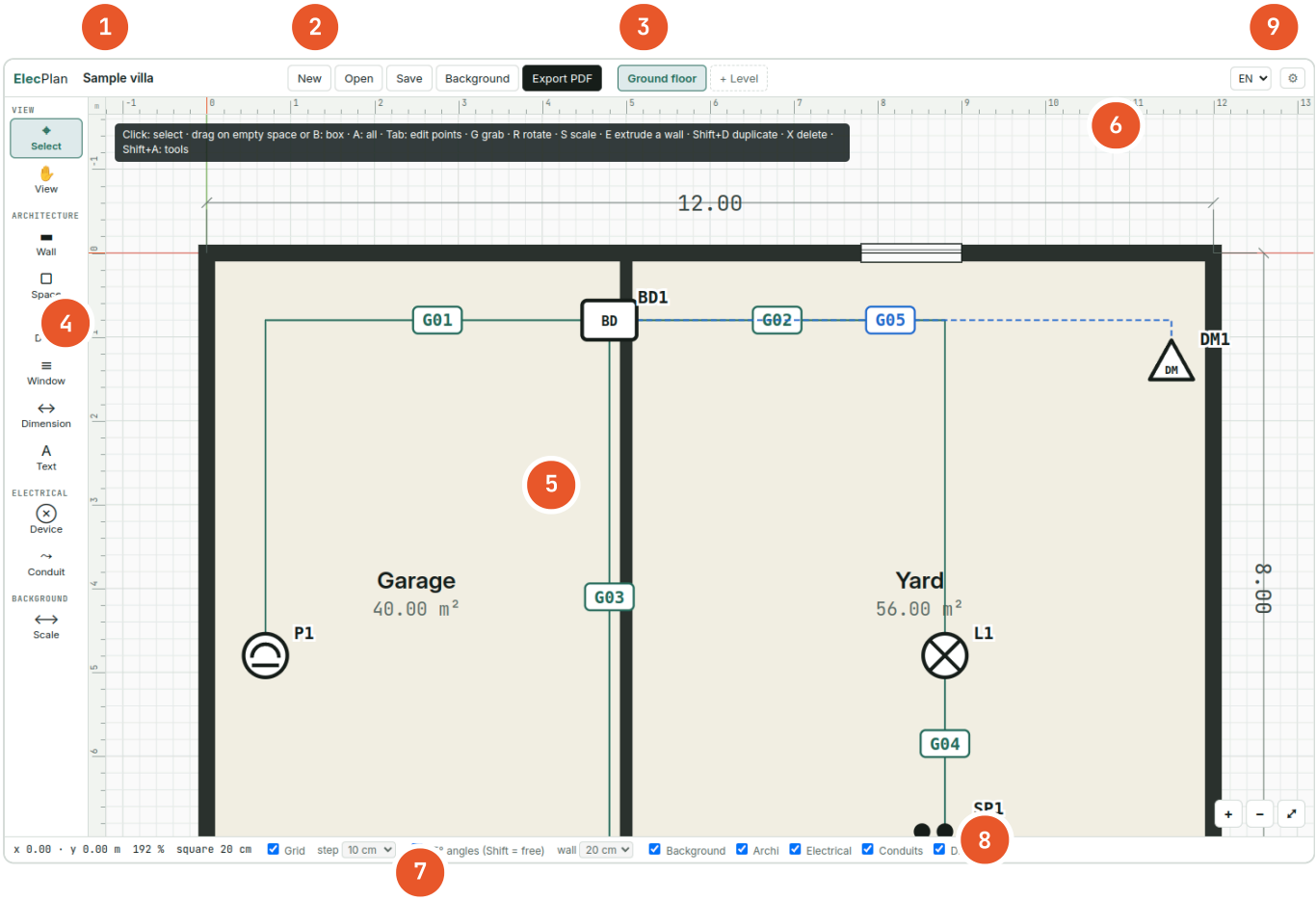
> AI tasks count towards your Claude plan's usage limits, just like using Claude Code on the command line.

Open the **AI assistant** tab to check: the "Claude Code" card should show ✅ and the detected version. Otherwise, see *Troubleshooting*.

Development

```
npm install
npm start        # run from the sources
npm run manual   # build this manual (PDF + HTML)
npm run dist     # build .AppImage and .deb (also builds the manual)
```

The main screen



	AREA	PURPOSE
1	Project name	Click to edit it. "● unsaved" means there are changes to save.
2	Files	New, Open, Save, Background, Export PDF.
3	Levels	One tab per level. Double-click to rename, "+ Level" to add one.
4	Tools	Select, view, architecture and electrical tools, and background scaling. `T` hides the toolbar.
5	Plan	Drawing area, with rulers in metres at the top and left (cursor in orange, selection in blue) and the X (red) and Y (green) axes through the origin. Middle-drag: pan; wheel or Ctrl + middle-drag: zoom. `Shift+A`: tools menu.
6	Panel	Properties of the selection, device library, bill of materials (Materials), AI assistant. `N` hides it.
7	Status bar	Coordinates, zoom, grid square size, grid and step, 15° angles, wall thickness, visible layers, messages.
8	Zoom	+, −, and ⤢ to fit everything.
9	Language and preferences	FR / EN switches the interface language immediately, without losing your work (also View → Language). On first launch, the language follows the system. ⚙ opens the preferences.

Preferences

⚙ at the top right, `Ctrl+,` or Edit → Preferences. They apply to all projects and are kept between sessions:

- **Colours** of the interface, the plan (background, walls, space fill, device symbols, grid, axes) and each conduit kind;
- **Display**: rulers, scene axes, grid lines, zoom speed;
- **Values at startup and for new items**: wall thickness, grid step, snapping, 15° angles, door and window widths, position of a space's walls, area shown on labels.

↺ next to a changed value restores its factory value; "Reset all to factory defaults" restores them all.

Getting started

1. **Create the project.** Click “New”, then rename the project at the top left. Rename the “Ground floor” level if needed (double-click its tab).

2. **Import a background (optional).** “Background” → image or PDF. Then the **Scale** tool: click 2 points whose distance you know and enter it in metres.

3. **Draw the walls.** The **Wall** tool `W`: click each corner. Click the first point to close, or double-click / `Enter` to finish. Pick the thickness in the status bar.

4. **Define the spaces.** The **Space** tool `R`: drag from one inside corner to the other. Name it and set its width and height in Properties; tick “Add walls” (or `E`) to build its walls if you have not drawn them.

5. **Place doors and windows.** Tools `D` and `F`: hover over a wall and the opening snaps to it. `R` rotates the leaf, `Shift+R` flips it.

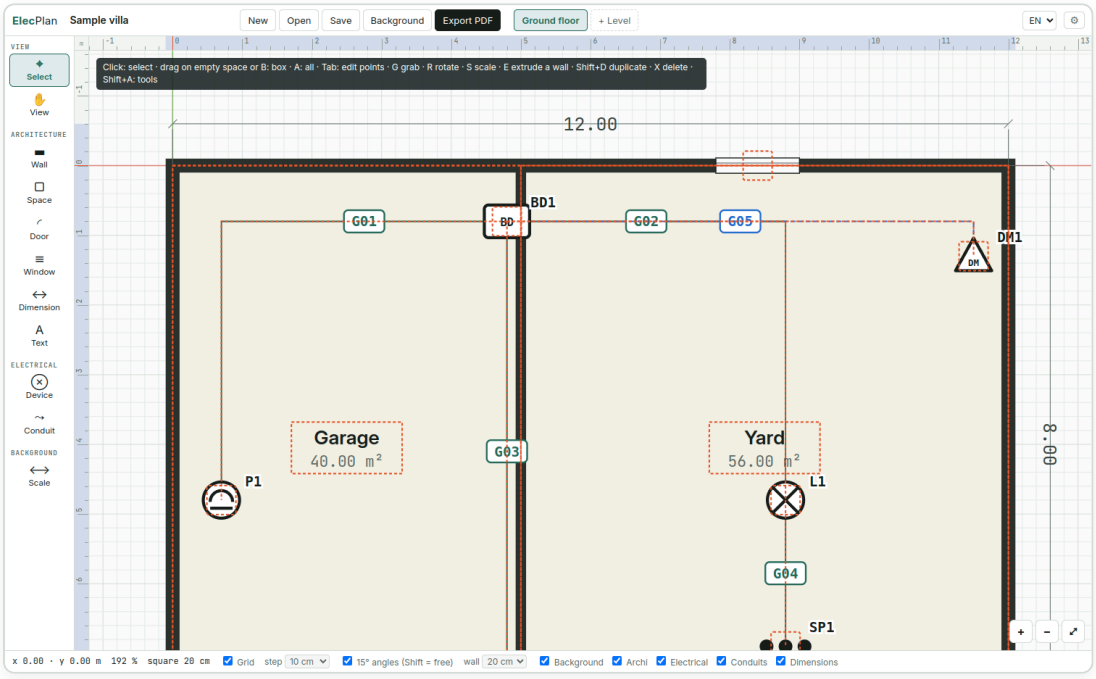
6. **Add dimensions.** The **Dimension** tool `C`: 2 clicks. Set the offset in Properties.

7. **Place the devices. Devices** tab: pick a type, then click on the plan. The reference is automatic. Fill in circuit and mode in Properties.

8. **Draw the conduits.** The **Conduit** tool `G`: click the start device (often a junction box), the bends, then the end device. Enter the content (R25.1, B25.1…).

9. **Check. Materials** tab for quantities and lengths; **AI assistant** tab → “Check compliance”.

10. **Save and export.** `Ctrl+S` to save the project, “Export PDF” for the A3 plan of each level.



Drawing a wall: the length is shown live next to the cursor.

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Architecture tools

TOOL	KEY	HOW TO USE	TIPS
Select	`V`	Click to select, drag to move. **Edit mode** (`Tab` or double-click) on a wall, space, conduit or dimension: click or box-select points, then G R S E X act on those points; `Tab` to go back.	**Several items:** drag on empty space or inside a space to draw a box (right: fully inside; left: touching). Shift+click adds or removes. `G` `R` `S` `E`, `Shift+D`, `X`, `A`, `B`: as in Blender (see *Keyboard shortcuts*). In Edit mode, double-click a segment: adds a point; a point: removes it. Arrows: move by the grid step (Shift: 50 cm).
View	—	Drag to pan the view.	At any time, as in Blender: middle-drag (or `Space` + drag). Ctrl + middle-drag or wheel: zoom at the cursor. Home: fit all.
Wall	`W`	Click point by point. To finish: double-click, `Enter`, right-click, or click the first point (closed wall).	Angles in 15° steps and lengths on the grid step; `Shift` for a free angle. Automatic snapping to existing vertices. **Exact length:** type it in metres (`3.4`), then `Enter`; `3.4<90` also sets the angle (0° = right, 90° = up).
Space	`R`	Drag from corner to corner (or click, then click): a rectangle.	Width and height in Properties while it stays rectangular (even rotated); other shapes in Edit mode. Creates no walls; see *Spaces* below.
Door	`D`	Hover over a wall, click to place (83 cm by default, see Preferences).	`R`: rotates the leaf; `Shift+R`: flips it. Snaps to the middle of the wall (between corners and joining walls) and shows the distances; "Centre on wall" in Properties.
Window	`F`	Hover over a wall, click to place (120 cm by default).	Aligns with the wall, takes its thickness and snaps to the middle, like the door.
Dimension	`C`	Click the 2 points to measure.	Positive or negative offset in Properties; the value follows the points.
Text	`Shift+Space` T	Click, then type.	Double-click with Select to edit.
Scale	—	2 clicks on the background, then the real distance in metres.	Do this before drawing over an imported background.

Spaces

- **Walls:** the "Add walls" checkbox (or `E`) puts an ordinary closed wall around the space. Set its thickness and position: outside (the area stays the one drawn), centred on the outline, or inside. Changing the size, thickness or position rebuilds it from the outline. Untick to remove it.
- **Label:** the name and area form a separate label. Drag it, or use G / R / S to move, rotate or resize it; double-click to rename the space. "Show area" hides or shows the area. The label follows the space when it is moved, rotated or duplicated.

Grid and snapping

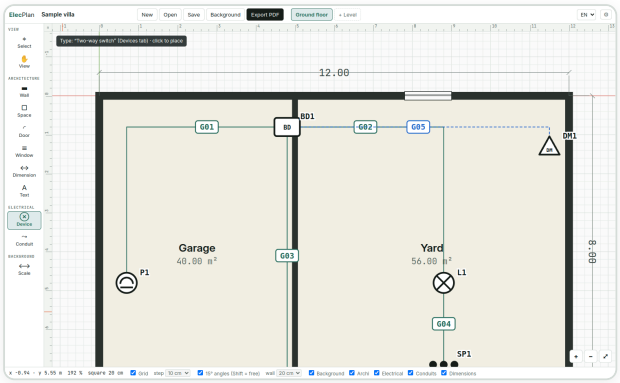
“Grid” checkbox and step (5 to 50 cm) in the status bar; hold `Alt` while moving to ignore it. Zoomed out, the squares grow to 20 cm, 50 cm, 1 m… (size shown in the status bar), with darker lines on whole metres. With G, R or S, hold `Ctrl` to snap: G brings a corner, a space centre or a device onto a nearby point of another item, or else onto the grid.

Layers

The Background, Archi, Electrical, Conduits and Dimensions checkboxes hide a layer on screen; a hidden item cannot be selected.

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Devices and circuits



Devices tab: pick a type, then click on the plan.

Placing a device

1. **Devices** tab, click the type.
2. The **Device** tool is activated: click on the plan.
3. The reference is assigned: P for sockets, I for switches, L for light fittings, SP for spotlights, BD for junction boxes, DM for motion detectors… (the prefixes come from the French names and are the same in both languages).

Device properties

FIELD	USE
Ref.	Editable; keep it unique
Type	Changes the symbol
Description	Free text (“Garage utility socket”)
Circuit	Chosen from the project circuits
Mode	S (single), VV (two-way), two-way master / slave, ON, DM, DM/ON, BP (push button)
Height	Mounting height in cm
Note	Site remark

`R` rotates the selected device (type `90` then Enter for a quarter turn). `Shift+D` duplicates it with a new reference. Deleting a device also deletes its conduits (a message says so).

Circuits

With nothing selected, the Properties panel lists the project circuits (name, rating, section, panel). “+ Circuit” adds one; they are offered in each device’s properties.

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Conduits

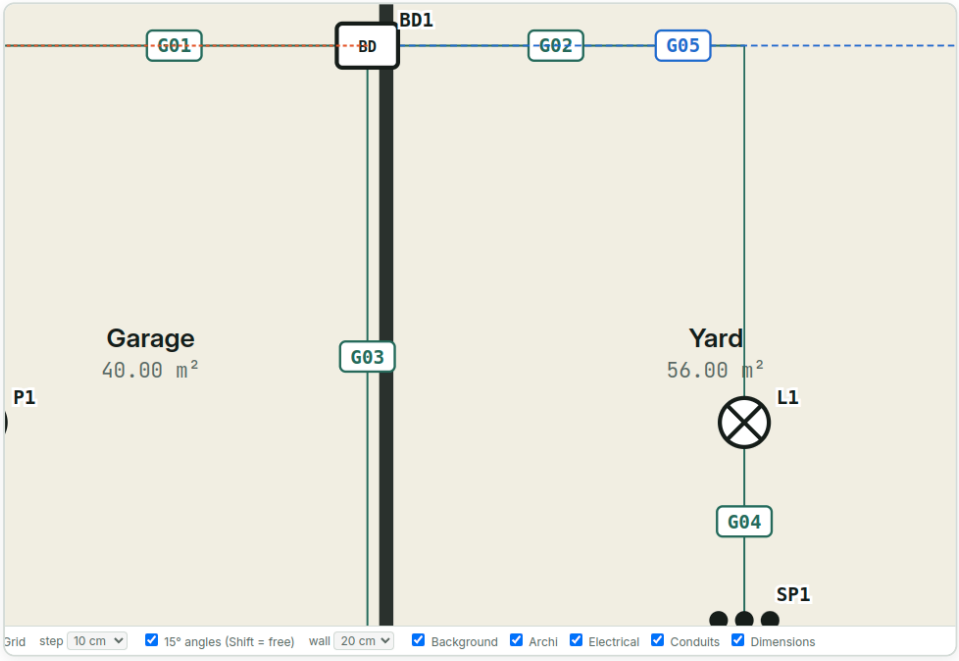
Drawing

1. The **Conduit** tool `G`.
2. Click the start device (panel, enclosure, junction box…).
3. Click for the bends.
4. Click the end device: the conduit is created with the next free number (G01, G02…).

`Esc` cancels the conduit being drawn. The ends follow the devices when they are moved.

Editing

- `Tab` (or double-click the conduit): Edit mode. Drag a bend to move it, or pick it and press G.
- Double-click a segment: adds a bend; double-click a bend: removes it.
- **Curve:** Properties → Path → "Curve (Bézier)". Each node gets two aligned round handles; drag them to bend the conduit. "Smooth" recalculates an even curve. The length shown is that of the curve.

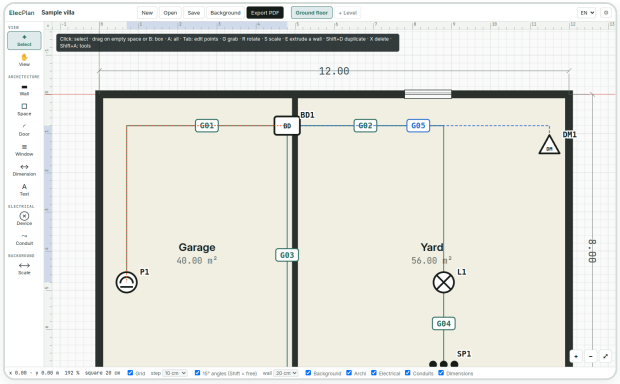


Conduit G01 in Edit mode (Tab): square handles on the bends.

Properties

FIELD	EXAMPLE
Number	G07 (editable)
Kind	Power · Low voltage · Spare · Outdoor / buried
Tube	ICTA Ø25
Content	R25.1, B25.1, T25.1

Recommended wire coding: letter (R phase, B neutral, T earth, J traveller, N switched return) + section (15 = 1.5 mm², 25 = 2.5 mm²) + network number. Same code from end to end, Wago connectors included.



Conduit properties.

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Background, levels and bill of materials

Background plan

“Background” imports an image (PNG, JPG, WEBP) or the first page of a PDF. It is placed 20 m wide by default: use the **Scale** tool to give it the right size. Opacity is set in Properties (with nothing selected); “Remove” deletes it.

Levels

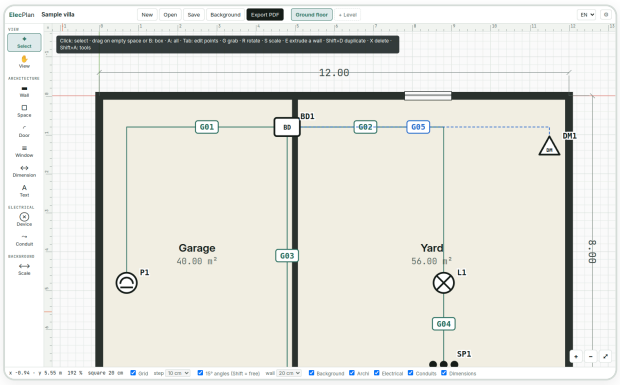
“+ Level” adds a level; double-click a tab to rename it. Each level has its own walls, devices, conduits and background. References and conduit numbers stay unique across the project. The PDF exports one level per page.

Bill of materials

The **Materials** tab gives:

- the number of devices per type and per level;
- the level’s conduits (start, end, content, length on plan);
- the spaces and their areas.

“Export bill of materials (CSV)” produces a file for any spreadsheet (comma separator, decimal point). In French, the file uses semicolons and decimal commas.



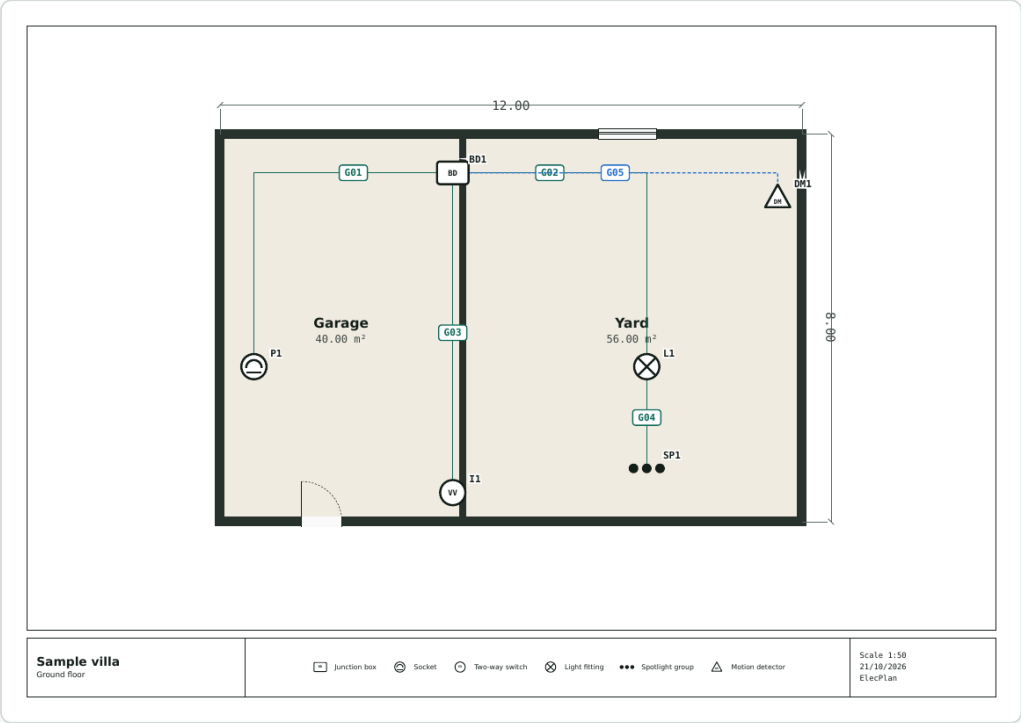
Materials: devices, conduits and spaces.

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Save, open, export

ACTION	SHORTCUT	RESULT
Save	`Ctrl+S`	Project file `.elecplan.json` (all data, background included)
Save as	`Ctrl+Shift+S`	New copy of the project
Open	`Ctrl+O`	Loads a project; offers to discard unsaved changes
Open Recent	File menu	The last 10 projects opened or saved; “Clear Recently Opened” empties the list
Export PDF	`Ctrl+P`	A3 landscape, one level per page, standard scale (1:20 to 1:1000), title block and legend of the symbols used
Export SVG	File menu	Vector plan of the current level, for drawing software
Bill of materials CSV	Materials tab	Devices and conduits of the whole project

Exports (PDF, CSV) use the interface language and your colour preferences at the time of export.



A3 PDF export: plan to scale, title block (project, level), legend, scale and date.

Recovery

Every change is kept in local storage. The application always starts with an empty project; after a crash, File → “Recover Unsaved Work” reloads the last work. This does not replace saving the project. Quitting with unsaved changes asks you to “Save”, “Don’t save” or “Cancel”.

Undo

`Ctrl+Z` undoes, `Ctrl+Shift+Z` or `Ctrl+Y` redoes (120 steps). An applied assistant proposal is also undone with a single `Ctrl+Z`.

Sharing a project

The `.elecplan.json` file is readable JSON text: it can be sent, archived or processed by a script (see *Project file format*).

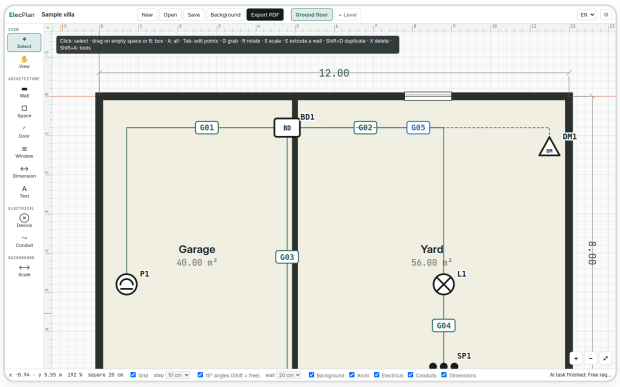
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AI assistant

How it works

1. You pick a task and start it.
2. ElecPlan creates a task folder with `CONSIGNE.md` (the instructions), `projet.json` and, if needed, the image.
3. It runs `claude -p` in that folder, in the background. The log is shown live; you keep drawing.
4. Claude writes `resultat.json` (the proposal) and `rapport.md` (the report).
5. "Show result", then "Apply": the proposal is added to the current level (undo with `Ctrl+Z`).

Tasks run one at a time, in order. "Cancel" stops a task; "Folder" opens its working folder. When the interface is in English, Claude writes its report and notes in English; the trade conventions sent to it stay in French.



Finished task: log, proposal to apply and report.

Security

- By default, Claude can only read and write files in the task folder; system commands are blocked.
- "Allow commands" (full-file task, or Settings) lets it produce a PDF, for example.
- The project sent does not include the background image (except the sketch task, which sends the chosen image).

The five tasks

TASK	INPUT	RESULT	TIPS
Read a sketch or photo	An image; details (orientation, known dimensions, scale)	Walls, spaces, openings, devices, conduits to apply; report of assumptions	Sharp photo, taken straight on. Give one real dimension: the result will be to scale.
Propose conduits and wiring	The plan with its devices; instructions (junction boxes to use, required circuits); replace or complete	Conduits drawn and numbered, coded content, circuits, assignments; report (fill, Wago per box)	Place the panel and junction boxes first. Without "Replace", existing conduits are kept.
Generate the full electrical file	The project; title, date, special points	dossier.html (and dossier.pdf if commands are allowed) in the task folder	If the "dossier-electrique-zone" skill is installed in Claude Code, its method is followed.

TASK	INPUT	RESULT	TIPS
Check compliance	The project; priority points	Sorted list (blocking, important, advice) with fixes; “Show” button to locate	Run it before the final export.
Free request	Your request as text	Answer in the report; any changes to apply	E.g. “Add an IP44 socket next to each outside door”.

Settings

In the AI assistant tab → Settings: command to run (`claude` by default, or a full path), model (empty = Claude Code’s default), permission to run commands for all tasks.

Usage and plan

Tasks use the account signed in to Claude Code and count towards the plan’s limits. A simple task takes a few seconds to a minute; a full electrical file can take several minutes.

Conventions sent to Claude

2-pole type C breakers, 30 mA RCDs, solid H07V-U wires, ICTA Ø25 conduit (Ø30 for charging points) filled to one third at most, 16 A / 20 A on 2.5 mm^2 and 6–10 A on 1.5 mm^2, R/B/T/J/N codes + section + number, conduits never renumbered, Wago connectors in junction boxes and wherever a wire splits, through-wires kept continuous, looping on double terminals. They are written in `src/ai.js` and can be changed.

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Keyboard shortcuts

Shortcuts follow Blender where it makes sense.

KEY	ACTION
`Middle-drag` / `Space` + drag	Pan the view
`Ctrl + middle-drag` / `Wheel`	Zoom
`Home` / `Numpad .`	Fit all / frame selection
`N` / `T`	Hide side panel / toolbar
`Shift+A` / `Shift+Space`	Tools and devices menu
`V W R D F C E G`	Nothing selected: select, wall, space, door, window, dimension, device, conduit
`Tab`	Object mode ↔ Edit mode (points of walls, spaces, conduits, dimensions)
`A` / `Alt+A` / `Ctrl+I`	Select all / deselect / invert
`B` / `Shift+click`	Box select / add, remove
`G` / `R` / `S`	Grab / rotate / scale: free; Ctrl snaps (nearby points or grid, 15°, 0.1); X / Y: axis; typed value
`E`	Extrude the selected wall; on a space: add its walls
`Shift+D`	Duplicate, then grab
`X` / `Del`	Delete the selection
`Enter` / `Esc`	Confirm / cancel (also left / right click)
`3.4` / `3.4<90`	Length (m) and angle while drawing a wall
`Arrows`	Move (Shift: 50 cm)
`Ctrl+Z` / `Ctrl+Shift+Z`	Undo / redo
`Ctrl+S` / `Ctrl+O`	Save / open
`Ctrl+,`	Preferences
`F12` / `Ctrl+P`	Export PDF
`Shift` **(drawing)** / `Alt` **(moving)**	Free angle / ignore grid

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Troubleshooting

PROBLEM	SOLUTION
“Claude Code not found”	Install it (`npm i -g @anthropic-ai/claude-code`). If it is installed elsewhere, put its full path in Settings (`which claude`).
The task fails immediately	Run `claude` in a terminal and do `/login`. Check the Internet connection.
“done (no output file)”	Claude wrote neither resultat.json nor rapport.md: run it again with a more precise request; check the log and the folder.
Usage limit reached	Wait for your plan to reset; drawing features remain available.
The AppImage won’t start	Install `libfuse2`, or use the .deb package.
The PDF background is blurry	It is rendered at 3×; for more detail, export the page as a high-resolution image and import that.
An item can’t be selected	Check that its layer is ticked in the status bar.
Symbols are too small	Properties (nothing selected) → Symbols → Size.
Colours or defaults look wrong	Preferences (⚙) → “Reset all to factory defaults”.
The interface is not in my language	FR / EN selector at the top right, or View → Language menu.

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Project file format

An `.elecplan.json` project is readable JSON (older `.fpp.json` projects open too). Units in centimetres, origin at the top left, Y pointing down. Spaces are stored under `rooms`, their labels under `labels`, conduits under `gaines`.

```
{
  "app": "ElecPlan", "version": 1, "name": "Sample villa",
  "settings": { "grid": 10, "symbol": 2 },
  "circuits": [ { "id": "c1", "name": "Lighting 1", "cal": "10A", "section": "1,5", "tableau":
"TD" } ],
  "levels": [ {
    "id": "lv…", "name": "Ground floor", "elevation": 0,
    "background": { "src": "…", "x": 0, "y": 0, "w": 2000, "h": 1400,
"opacity": 0.5 },
    "walls":    [ { "id": "w…", "pts": [[0,0],[1200,0],[1200,800]], "thick": 20 } ],
    "rooms":    [ { "id": "r…", "name": "Garage", "pts": [[0,0],[500,0],[500,800],[0,800]] } ],
    "labels":   [ { "id": "l…", "space": "r…", "x": 250, "y": 400, "rot": 0, "size": 1, "area":
true } ],
    "openings": [ { "id": "o…", "type": "door", "x": 250, "y": 800, "angle": 0, "w": 83,
"flip": false } ],
    "dims":     [ { "id": "d…", "a": [0,0], "b": [1200,0], "offset": -60 } ],
    "texts":    [ { "id": "t…", "x": 100, "y": 100, "text": "Recess", "size": 18 } ],
    "devices":  [ { "id": "e…", "type": "prise_ip", "ref": "P1", "x": 60, "y": 450, "rot": 0,
                    "label": "IP44 socket", "circuit": "c2", "mode": "", "height": "30",
"note": "" } ],
    "gaines":   [ { "id": "g…", "num": "G01", "a": "e…(BD1)", "b": "e…(P1)", "pts": [[60,70]],
                    "kind": "pwr", "tube": "ICTA Ø25", "content": "R25.1, B25.1, T25.1" } ]
  } ]
}
```

A wall built for a space also carries `"space": "r…"` and `"spacePos": "out" | "mid" | "in"`. A curved conduit carries `"curve": true` and its handles in `"h"`.

Device types — prise, prise_ip, prise_spe, inter_s, inter_vv, inter_on, poussoir, dm, lum, applique, spots, bd, tableau, coffret, equip, rj45, reserve. The keys are French and identical in both languages (prise = socket, inter = switch, poussoir = push button, lum = light fitting, bd = junction box, tableau = panel, coffret = enclosure, reserve = spare).

Conduit kinds — pwr (power), lv (low voltage), res (spare), ext (outdoor / buried). Each conduit links two devices by their `id`; `pts` holds only the bends.

Planned for future versions

DXF import, single-line diagram, automatic space dimensioning, custom layers.